

inverse", *class="card-warning*", *class="card-danger*" and *class="card-success*".

- Spacing utilities, tooltip auto-placement support and the global font-size has been increased from 14px to 16px.
- The primary CSS unit is now *rem* rather than *px*. However, pixels are widely used for media queries.
- In Bootstrap v4, the Glyphicons font has been dropped. Suggested options are *fontAwesome* and *Octions*.

Angular

Angular is the right choice for creating a single page application. It helps to publish the Web component that is used in HTML pages. The latest version is Angular v6. In this version, the new features available include the component development kit, Angular CLI and an Angular Material Package update. In Angular, we use the Typescript language, which is the superset of JavaScript.

Installing Angular 6

Before installing Angular, we first need to check that Node.js is present in the machine. If so, then check which version it is, as well as the NPM version by using the following command:

```
node -v and npm -v
```

After that, execute the following command to install Angular:

```
npm install -g @angular/cli
```

Check the Angular version by using the command:

```
ng --version
```

To create a new Angular project, use the following command:

```
ng new my-first-app
```

Go to the *my-first-app* folder and run the *ng serve* command.

The basic structure of an Angular project

- *package.json*: This has the list of the node dependencies that are needed.
- *src/styles.css*: Global CSS is available throughout the application.
- *src/main.ts*: This is the main file that starts the Angular. The extension *.ts* stands for TypeScript.
- *src/index.html*: This is the first file that executes alongside *main.ts* when the page loads.
- *src/app/app.module.ts*: This is the file in which all the components, providers and modules are defined. Without defining them here, they can't be used elsewhere in the code.
- *src/app/app.component.html*: This is the main component of an Angular app, and all other components are usually

present within it.

- *src/app/app.component.ts* is the logic for this component.
- *src/app/app.component.scss* is the CSS for this component, which itself does not have important logic, but acts as a container for other components.

Why use Angular 6?

Some important reasons for using Angular are listed below.

- *Typescript 2.7+ support*: Since Angular v6 uses the latest version of Typescript, it is much easier to write code with the conditional type declaration, strict class initialisation and default declaration.
- *Elements*: Creating widgets or components which can include any existing page, was not a simple task in previous versions of Angular. Elements enable the creation of components, which will be published as Web components and can be used in any HTML page.
- *Service worker support*: In Angular, service workers are the scripts that run in the browser and manage the app cache. The latest version of Angular supports the navigation URL within the service worker, always remains in the current state (unless the server is reconnected) and updates the work.
- *i18n*: This is the important change in Angular 6, specially used for international languages. In i18n, with the help of pipe functionality, we can change the currency value with two digits. The i18n function is *getNumberOfCurrencyDigits()*. Some others are *formatNumber*, *formatCurrency*, *formatPercent* and *formatDate*.

There are many other important changes in Angular 6 such as the latest RxJS 6.0, tree shaking, the Bazel compiler, *ngModalChange*, etc.

ReactJS/ReactNative

ReactJS is a JavaScript library that supports both the front-end and the server for the user interface and Web applications.

React Native is a JavaScript mobile framework, which is used to compile native app components for mobile application development; it has support for different platforms (Android and iOS). Both have been open sourced by Facebook.

ReactNative and ReactJS both use ReduxJS. In both, components can be reused to save time for development. In React, when rendering a page completely from the server to the browser, the SEO of the Web app improves. It also improves the debugging speed, making the developer's life easier. If you are not familiar with React, don't worry, since it is easy to understand.

Differences between ReactJS and ReactNative

Some important differences between ReactJS and ReactNative are listed below.

- ReactNative is faster than ReactJS because when the latter is used for a new project, a bundler like Webpack is used to find which bundling module is needed for the project. But ReactNative comes with everything and is easy to set up.